MILITARY SPECIFICATION

BOOSTER, FUZE, MARK 60 MOD 2

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 60 MOD 2 Fuze Booster, referred to herein as "the booster."

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for.
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents and drawings. The following other Government documents and drawings of the issue specified in the solicitation form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

DEPARTMENT OF DEFENSE (DoD)

DoD 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND (Code Ident 30003)

DL 1137AS2150	Booster, Fuze, MARK 60 MOD 2.
SA 1137AS2160	Fixture, Cook-off Test.
SA 2747612	Fixture, Vibration, Three Plane.
SA 2747613	Fixture, Jolt Test.
SA 2747614	Fixture, 40-Foot Drop.

(Copies of drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-108	Steel Bars, Carbon, Cold-finished, Standard Quality, Specification for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The booster initiates the warhead of a guided missile (see 6.1) and is composed of a housing, cap, plastic cup, snap ring, wave spring, identification plate, and an explosive pellet containing approximately 84 grams of Composition CH-6 explosive (see Drawing 1137AS2150).

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3.1.1 Government-furnished property. Unless otherwise specified in the contract or purchase order, the Government will furnish the contractor with the following in a quantity sufficient to meet contract requirements (see 6.2.3):

a. Composition CH-6 explosive in accordance with MIL-C-21723.

b. MARK 9 MOD 0 Explosive Leads in accordance with LD 480141 for performance of the function and output test of 4.6.1.

c. Exploding bridgewire (EBW) detonators in accordance with 4.6.1.1.2.a for performance of the function and output test of 4.6.1.

3.2 Performance characteristics.

3.2.1 Function and output. The booster shall function on the first and only trial when detonated in accordance with 4.6.1 and Figures 1 and 2. The output shall produce a depth of dent of 0.140 inch minimum in the steel dent block of Figure 3.

3.2.2 Radiographic. The booster shall meet the following requirements (see 4.6.2):

a. There shall be no evidence of loose powder or other foreign material.

b. There shall be no evidence of voids, fissures, or cracks in the explosive pellet.

c. All parts shall be properly located as specified on the applicable drawings.

3.2.3 Environmental conditions. The booster shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the following environments.

3.2.3.1 Transportation vibration-temperature. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall meet the requirements of 3.2.1 after exposure to the environment of 4.6.3.1.

3.2.3.2 Temperature and humidity. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall meet the requirements of 3.2.1 after exposure to the environment of 4.6.3.2.

3.2.3.3 Random vibration. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall meet the requirements of 3.2.1 after exposure to the environment of 4.6.3.3.

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3.2.4 Safety conditions.

3.2.4.1 Twelve-meter drop. The booster shall not detonate nor burn during, and shall be safe (see 6.3) to handle and dispose of after exposure to the environment of 4.6.4.1.

3.2.4.2 Jolt. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall be safe (see 6.3) to handle and dispose of after exposure to the environment of 4.6.4.2.

3.2.4.3 Cook-off. The booster shall not detonate during exposure to the environment of 4.6.4.3. Any booster reaction (deflagration, burning, etc.) other than detonation is acceptable and may occur at any time.

3.2.5 Visual. The booster shall show no external evidence of cracks, burrs, dents, loose powder, or other deformities which may interfere with functioning or fit at the next assembly (see 4.6.5).

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspections (see 4.2 and 6.2.4).

3.4 Design and construction.

3.4.1 Production drawings. The booster shall meet all the requirements of and shall be manufactured and assembled in accordance with the drawings listed on DL 1137AS2150 and all associated documents.

3.4.2 Safety precautions. As a minimum, the safety precautions of DoD 4145.26M shall apply during manufacture, handling, and inspection of the booster, but shall not be considered as a substitution for the contractor's own safety procedures developed to accommodate the hazards associated with the booster.

3.4.3 Standards of manufacture.

3.4.3.1 Product cleanliness. No oils or solid lubricants of any kind shall be used on any part of the booster after final cleaning of components, except as specified on the drawings or herein.

3.4.3.2 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.4.3.3 Workmanship. Workmanship of the booster shall meet the requirements of MIL-A-2550.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsbility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein (see 6.2.1). Except as

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otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

4.2 First article inspections. Unless otherwise specified in the contract or purchase order, first article inspections shall be performed by the contractor and an independent test facility, after award of contract and prior to production (see 6.2.4). As specified in the contract or purchase order, first article inspections shall be performed on 30 boosters (see 3.3 and 6.2.4) which have been produced with the equipment, material, and procedures to be used in production. First article approval is valid only on the contract under which it is granted unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

4.2.1 First article inspection sequence. The first article of 30 boosters shall be subjected to the inspections of Figure 4 in the sequence shown. Inspections below the dashed line of Figure 4 shall be performed at the independent test facility designated in the contract or purchase order (see 6.2.4). At the conclusion of the inspection sequence and after receipt of first article approval, any remains of the first article shall be disposed of at the discretion of the cognizant demolition officer.

4.2.2 First article acceptance and rejection criteria. Each first article booster shall meet the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 4, the applicable performance requirements of Section 3 shall be met on the first and only trial; one or more failures shall be cause for refusal to grant first article approval. For the function and output test of 4.6.1, a depth of dent of less than 0.140 inch from any first article booster shall be cause for rejection of the first article. When computed in accordance with 4.6.1.3.d, the first article shall be accepted if the quantity $(\bar{X}-L)/s$ is equal to or greater than MIL-STD-414, Table B-1, k value, Acceptable Quality Level (AQL) of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the first article.

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4.3 Quality conformance inspections.

4.3.1 Inspection lot. As specified in the contract or purchase order, an inspection lot shall consist of completed boosters, fabricated under the same conditions and offered for acceptance inspection at the same time.

4.3.1.1 Inspection lot size. The inspection lot size shall be as specified in the contract or purchase order (see 6.2.1).

4.3.1.2 Inspection lot acceptance inspection sequence. After the inspection lot has been formed and identified by lot number, the lot shall be subjected to the inspections of Figure 5 in the sequence shown. When specified during the inspection sequence, a quantity of boosters shall be sampled from the inspection lot in accordance with MIL-STD-414, Inspection Level IV, Standard Deviation Method, Variability Unknown. The sample shall be shipped to the independent test facility specified in the contract or purchase order for performance of the inspections below the dashed line of Figure 5. The remainder of the lot shall be held in bond pending acceptance of the sample. If the sample does not cause lot rejection (see 4.3.1.3.b), the remainder of the lot shall continue through the inspection sequence as shown on Figure 5. Upon completion of the inspection sequence, the remains of the sample boosters shall be disposed of at the discretion of the cognizant demolition officer.

4.3.1.3 Inspection lot acceptance and rejection criteria. Each booster of the inspection lot shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 5, inspection lot acceptance or rejection shall be based on the following criteria:

a. For the inspections above the dashed line where 100 percent inspection is required, acceptance or rejection shall be on an individual basis. The failure of any booster to meet the applicable requirements of Sections 3 and 5 shall cause rejection of the booster.

b. For the inspections below the dashed line where sampling is required, each sample booster shall meet the applicable requirements of Section 3. For the function and output test of 4.6.1, a depth of dent of less than 0.140 inch from any sample booster shall be cause for rejection of the inspection lot. When computed in accordance with 4.6.1.3.d, the inspection lot shall be accepted if the quantity $(\overline{X}-L)/s$ is equal to or greater than MIL-STD-414, Table B-1, k value, AQL of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the inspection lot.

4.4 Measuring and test equipment. All measuring and test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a

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Calibration System Description in accordance with MIL-STD-45662 and the Data Item Description (DID) cited in 6.2.2. The Calibration System Description shall be accepted by the Government prior to use.

4.4.1 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions, and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured.

4.4.2 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein, are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.4 and 4.4.1, and are subject to approval by the procuring activity.

4.4.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.5 Test conditions and provisions. The following test conditions and provisions shall apply during performance of the inspection methods of 4.6:

a. Test environment. Unless otherwise specified herein or on the drawings, the test environment shall conform to 3.4.3.2.

b. Test plan. When specified in the contract or purchase order, the contractor shall prepare a test plan for the inspections specified herein. The test plan shall be prepared in accordance with the DID cited in 6.2.2 and MIL-T-18303, and shall be accepted by the Government prior to use. The contractor shall not deviate from the accepted test plan unless authorized by the Government.

c. Inspection record. When specified in the contract or purchase order, the contractor shall prepare and fill in an inspection record for each booster produced under a specified contract. The inspection record shall be prepared in accordance with the DID cited in 6.2.2. The results of all inspections specified herein, which may be extracted from the inspection and acceptance records maintained by the contractor, shall be recorded on the inspection record and shall be traceable to a specific lot.

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d. Inspection sequences. The inspections of 4.6 shall be performed in the sequences shown on Figures 4 and 5, as applicable. When feasible, sequential inspections may be combined provided it can be demonstrated by the contractor that the applicable requirements of Sections 3 and 5 have been individually met. The contractor shall not deviate from the inspection methods of 4.6 and additional tests shall not be added unless authorized by the Government.

4.6 Inspection methods. To determine conformance to the requirements of Sections 3 and 5, the following inspections shall be performed in accordance with the test conditions and provisions of 4.5.

WARNING

The booster contains explosive components which are sensitive and hazardous. Suitable safety precautions shall be observed during handling and testing of the booster (see 3.4.2).

4.6.1 Function and output. The function and output test consists of initiating the booster on a steel dent block (see 4.6.1.1.1.d) and measuring the depth of the dent to determine the output. The test operation and selection of test equipment and materials, except for those listed in 4.6.1.1.1, shall be the responsibility of the contractor. A list of recommended test equipment and materials is given in 4.6.1.1.2 to facilitate test setup by the contractor. To determine conformance to 3.2.1, test the booster in accordance with 4.6.1.2; then measure the depth of the dent and perform the computation as specified in 4.6.1.3.

4.6.1.1 Function and output test equipment and materials.

4.6.1.1.1 Required test equipment and materials. The following test equipment and materials are required to perform the function and output test:

a. Lead holder in accordance with Figure 2.

b. MARK 9 MOD 0 Explosive Lead (see 3.1.1).

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c. Detonator (an electric EBW detonator is recommended; see 3.1.1 and 4.6.1.1.2.a).

NOTE: Unless otherwise specified in the contract or purchase order, the Government will supply the contractor with EBW detonators (see 3.1.1). A contractor-selected detonator may be used subject to the approval of the output characteristics by the procuring activity. Approval of the contractor-selected detonator does not imply approval of the detonator's safety characteristics.

d. Dent block in accordance with Figure 3.

e. Expendable support blocks in accordance with Figure 3 to sustain the dent block of d. above during the measurement procedure of 4.6.1.3.

f. Dial indicator height gage and probe as specified in MIL-STD-331, Test 301.

g. Detonator firing source, low current resistance measurement device with the current limited to 10 milliamperes (mA) maximum, and interconnecting cables for the detonator of c. above.

4.6.1.1.2 Recommended test equipment and materials. The following test equipment and materials are recommended for safe performance of the function and output test in conjunction with suitable safety procedures and facilities:

a. EBW detonator of 0.190 ±0.001 inch diameter with an output charge of 70 ±10 milligrams of pentaerythritol tetranitrate (PETN) pressed at 10,000 ±500 pounds per square inch (lb/in^2) (see 3.1.1).

b. Firing source and cables for the detonator of a. above.

c. Firing bay or barricades for the blast and fragment hazards of the booster.

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4.6.1.2 Function and output test method. Test the booster as specified below using the test setup shown on Figure 1:

WARNING

The function and output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 3.4.2).

a. Place the steel dent block of Figure 3 on the steel plate, as shown on Figure 1.

b. Place the booster in the center of the dent block, booster cap up.

c. Place the loaded lead holder, without the detonator, on the booster as shown on Figure 1.

NOTE 1: The installation and wiring of the detonator shall be performed in accordance with the contractor's safety procedures (see step h. below).

NOTE 2: Steps d. through l. below are suggested as good safety procedures appropriate for the hazards involved.

d. Clear the firing area of all nonessential personnel and equipment.

e. Verify that the detonator leads are shorted and that ordnance personnel, the firing cables, and the booster assembly are grounded.

f. Verify that the firing cables are not connected to the firing source.

g. Insert the detonator into a hand-safe containment block, remove the short from the detonator leads, connect the detonator leads to the firing cable, and insulate the connections.

h. Insert the detonator into the lead holder so the detonator is bottomed against the explosive lead, as shown on Figure 1.

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i. Remove all remaining personnel and nonexpendable equipment from the test bay and verify that all openings through which fragments could escape are secured.

WARNING

The booster causes a significant blast and fragment hazard. Verify that the area is cleared of all personnel before applying energy to the firing circuit (see 3.4.2).

j. Connect the firing cable to the firing source.

k. Verify that all personnel are clear, sound the firing alarm, and initiate the detonator.

l. Allow sufficient time to elapse for the fumes to clear the firing bay, resecure the firing source, disconnect and short the firing cables, and retrieve the dent block.

4.6.1.3 Output dent measurement procedure. Measure the depth of dent as follows:

a. Place the dent block on the support blocks of Figure 3.

b. As shown on Figure 3, place the gage probe of the dial-indicator height gage (see 4.6.1.1.1.f) in the deepest part of the dent within the shaded area. Zero the gage indicator.

NOTE: Do not change the relative position of the dent block on the support blocks after zeroing the gage indicator.

c. Remove the probe from the dent and take gage readings at two points, each approximately 0.125 ±0.030 inch from the opposite edge of the dent block, as shown on Figure 3. Average the two readings to determine the depth of dent and conformance to 3.2.1 (see 4.2.2 and 4.3.1.3.b).

d. Using the data from this test, compute the quantity $(\bar{X}-L)/s$, as defined in MIL-STD-414, Single Specification Limit, Form 1, Variability Unknown, Standard Deviation Method. The lower limit, L, of the booster output to be used in the computation shall be 0.145 inch (see 4.2.2 and 4.3.1.3.b).

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4.6.2 Radiographic. To determine conformance to 3.2.2, perform a radiographic inspection in accordance with MIL-STD-453 of each booster using radiographic equipment capable of detecting assembly defects such as cracks, voids, fissures, or loose powder. Take a minimum of two X-rays of each booster, each 90 ±5 degrees apart and showing the longitudinal section through the booster (see Drawing 1137AS2150).

4.6.3 Environmental tests. After exposure to the following environments, the booster shall be subjected to the test of 4.6.1 in the sequence shown on Figure 4 or 5, as applicable.

4.6.3.1 Transportation vibration-temperature. To determine conformance to 3.2.3.1, test the booster as follows:

a. Mount the vibration fixture of SA 2747612 on a vibration table and tighten the hold-down bolts to a torque of 20 ±1 foot-pounds (ft-lb).

b. Mount the booster to the vibration fixture and tighten the mounting screws to a torque of 17 ±1 inch-pounds (in-lb).

c. Subject the booster to the test of MIL-STD-331, Test 119, Procedure 2, except first article boosters shall be temperature-conditioned to and vibrated at the following temperatures (see Figure 4):

(1) Ten boosters at +160 ±5 degrees Fahrenheit (°F).

(2) Ten boosters at -65 ±5°F.

4.6.3.2 Temperature and humidity. To determine conformance to 3.2.3.2, test the booster in accordance with MIL-STD-331, Test 105, Two Chamber Method.

4.6.3.3 Random vibration. To determine conformance to 3.2.3.3, test the booster as follows:

a. Repeat 4.6.3.1.a and b.

b. Subject the booster to the test of MIL-STD-810, Method 514.2, Equipment Category d.1, Procedure IIA, Table 514.2-IV, Figure 514.2-4, except first article boosters shall be temperature-conditioned to and vibrated at the following temperatures (see Figure 4):

(1) Ten boosters at +160 ±5°F.

(2) Ten boosters at -65 ±5°F.

NOTE: To prevent damage to test equipment, limiting of instantaneous acceleration peaks to three times the overall root-mean-square acceleration level is permitted.

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4.6.4 Safety tests.

4.6.4.1 Twelve-meter drop. To determine conformance to 3.2.4.1, test the booster as follows:

a. Mount the booster to the drop test fixture of SA 2747614 and tighten the hold-down screws to a torque of 17 ±1 in-lb.

b. Subject the booster to the test of MIL-STD-331, Test 103, Procedure I.

NOTE: Breakage of the explosive pellet and the plastic cup, or the presence of loose pieces of explosive in the drop test fixture shall not be considered criteria for failing the test.

4.6.4.2 Jolt. To determine conformance to 3.2.4.2, test the booster as follows:

a. Mount the jolt fixture of SA 2747613 to the jolt machine and tighten to a torque that will prevent loosening during the test.

b. Mount the booster to the jolt fixture and tighten the hold-down screws to a torque of 17 ±1 in-lb.

c. Subject the booster to the test of MIL-STD-331, Test 101, Procedure II.

4.6.4.3 Cook-off. To determine conformance to 3.2.4.3, test the booster as follows:

WARNING

The cook-off test involves hazardous operations. Suitable safety precautions shall be observed (see 3.4.2).

a. Install the booster in the cook-off test fixture of Drawing SA 1137AS2160 as specified on the drawing and as shown on Figure 6.

NOTE: Use of a high temperature anti-seize compound between all threaded surfaces will greatly facilitate disassembly after the test.

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b. Position the assembly of a. above over a gas burner array as shown on Figure 7. The height of the assembly above the array shall be such that the assembly will be engulfed in flame.

NOTE: Unless otherwise specified in the contract or purchase order, the test setup of b. above shall be approved by the procuring activity prior to test (see 6.2.1).

WARNING

During the direct flame impingement of c. below, some portions of the cook-off test fixture fuze simulator (see Figure 6) may be expelled violently.

c. Subject the assembly to direct flame impingement at an average flame temperature of 1,500°F until all ordnance reactions are completed or until 15 minutes maximum have elapsed. Continuously monitor and record the temperature of the flame at the two locations indicated on Figure 7.

4.6.5 Visual. To determine conformance to 3.2.5 and the general requirements of Section 3, visually inspect the booster.

4.6.6 Packing. To determine conformance to Section 5, inspect the packaging, packing, and marking of the booster prior to shipment.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified in 5.1.1.

5.1.1 Level C packaging. Boosters shall be packaged separately and in a manner that will provide adequate safety and protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified in 5.2.1.

5.2.1 Level C packing. Boosters, packaged in accordance with 5.1, shall be packed in a manner that will provide adequate safety and protection from damage during direct shipment from the supply source to the first receiving activity. Packing shall be suitable for explosives and shall conform to 49 CFR 171-178.

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5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The booster is intended for use in the AIM-54C Phoenix Guided Missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

c. Inspection lot size (see 4.3.1.1).

d. Independent test facility to perform the inspection lot acceptance inspections below the dashed line of Figure 5 (see 4.3.1.2).

e. Government final inspection gage drawings, if required (see 4.4.3).

f. Approval by procuring activity of cook-off test setup, if required (see Note, 4.6.4.3.b).

g. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should

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be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Options
4.4	Calibration System Description	DI-R-7064	---
4.5.b	Test Plan	DI-T-5204	---
4.5.c	Inspection Record	UDI-M-20404	---

(DIDs related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

6.2.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 30 boosters, as specified in 4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article. In addition, the contracting officer should identify the independent test facility to perform the inspections below the dashed line of Figure 4 (see 4.2.1).

6.3 Definition of safe. Safe is defined as the ability of the booster to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For non-service use, safe is defined as the ability of the booster to maintain its safety features in a condition which will permit disposal without injury to personnel.

Preparing activity:
Navy - AS
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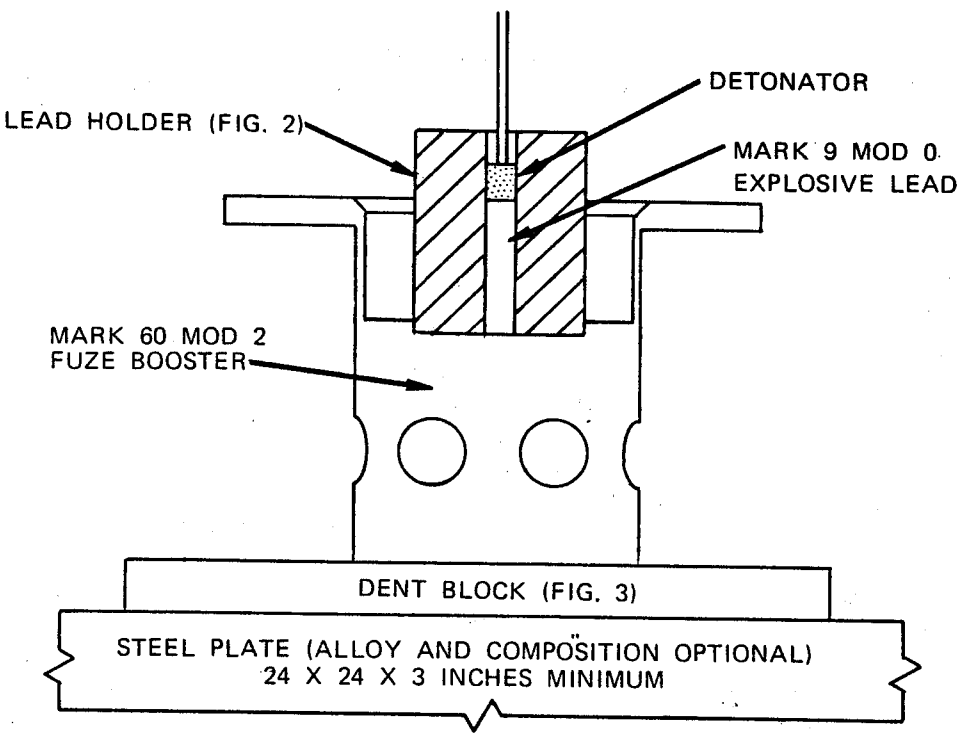
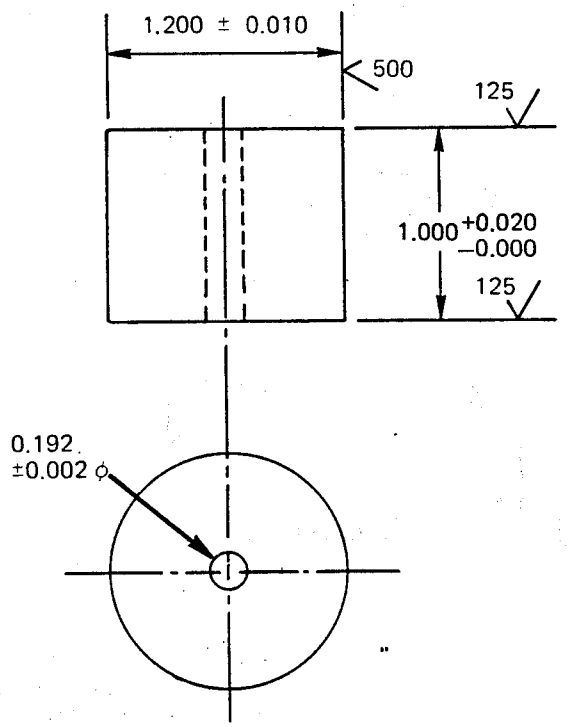


FIGURE 1. Function and output test setup.

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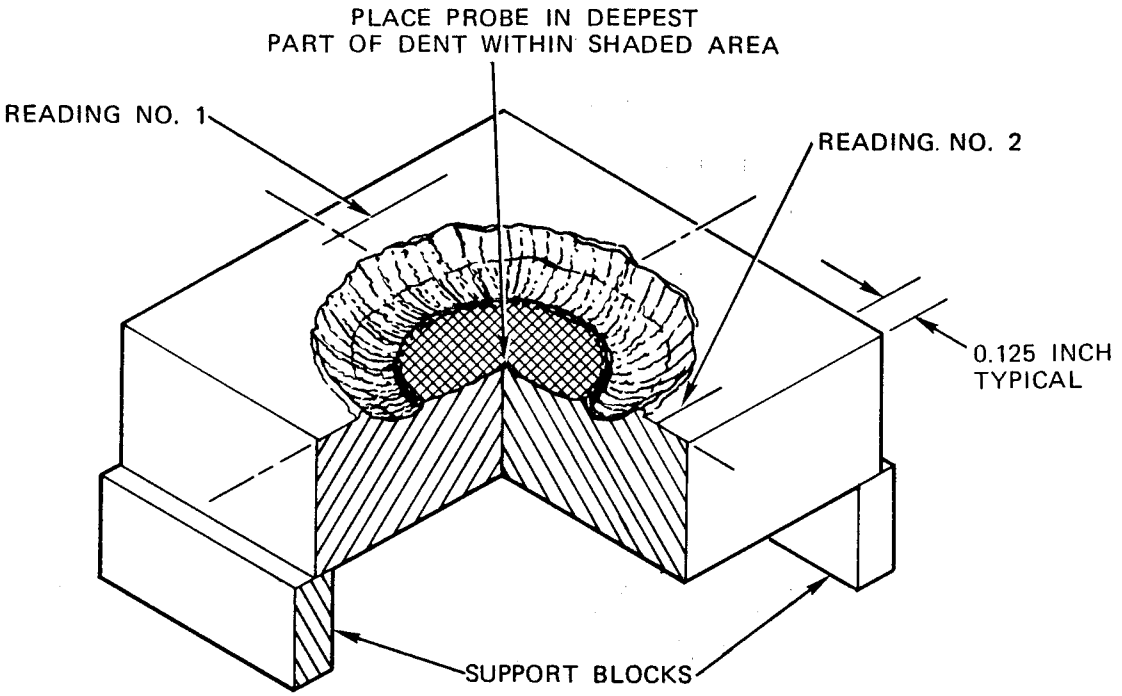
LEAD HOLDER

NOTES:

1. Material shall be steel, ASTM A–108, Grade 1018 through 1035, cold– or hot–rolled.
2. Dimensions are in inches.

FIGURE 2. Lead holder for function and output test.

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NOTES:

1. Dent Block: Steel, cold finished bar in accordance with ASTM A–108, Grade 1018 or 1020, Rockwell hardness B70 to B95, surface condition as rolled. Size 5.000 ±0.125 inches square, 1.5 inch stock. Upper and lower surfaces of block shall be clean and free from rust and burrs.
2. Support Blocks: Material and size shall be at the contractor's discretion.

FIGURE 3. Dent block and output dent measurement for function and output test.

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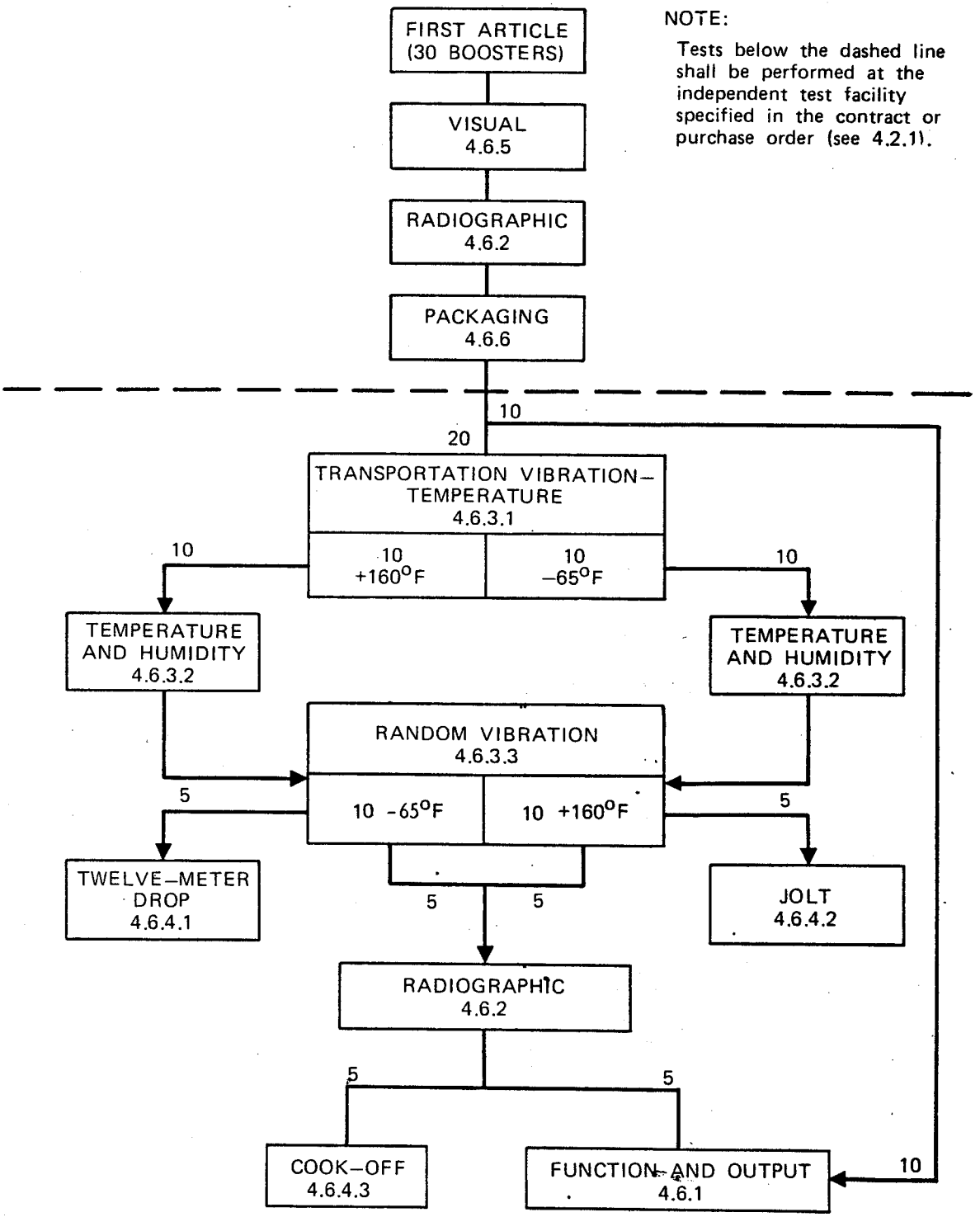


FIGURE 4. First article acceptance inspection sequence.

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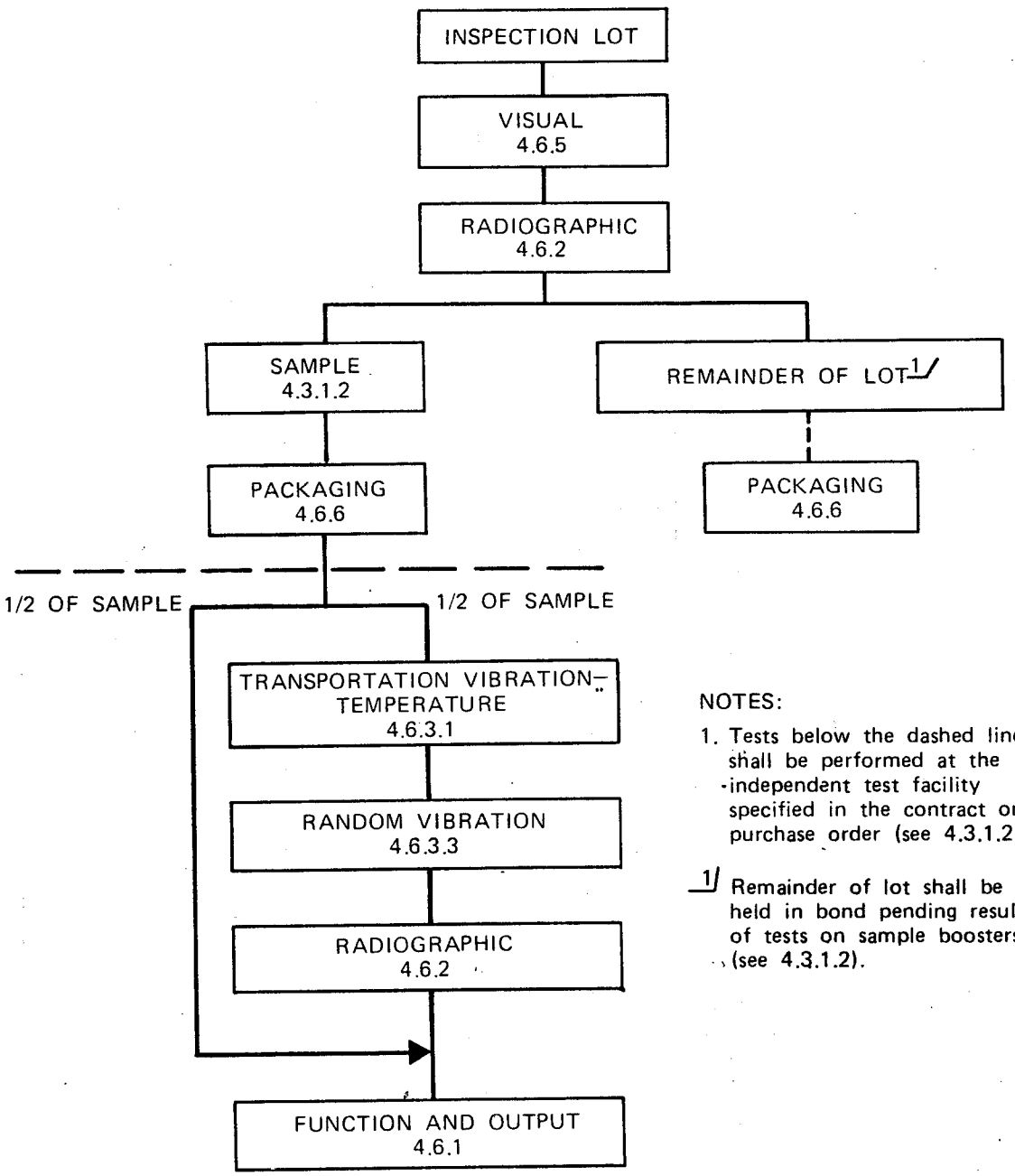
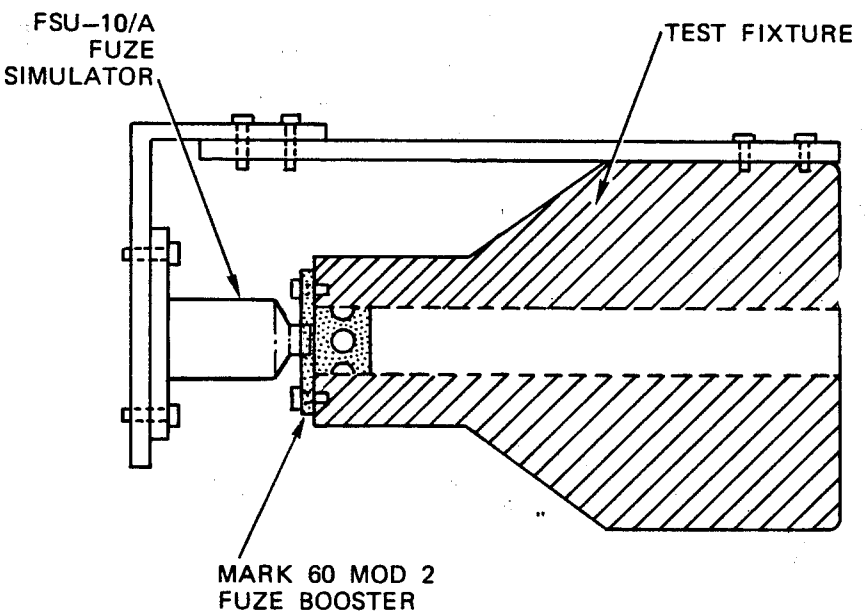


FIGURE 5. Inspection lot acceptance inspection sequence.

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NOTES:

1. There shall be a 0.040 ±0.010 inch air gap between the nose of the fuze simulator and the plastic booster end-cap.
2. Fixture and fuze simulator are in accordance with Drawing SA 1137AS2160. Fixture is designed to simulate the thermal response of the MARK 82 MOD 0 Warhead.

FIGURE 6. Booster installed in test fixture for cook-off test.

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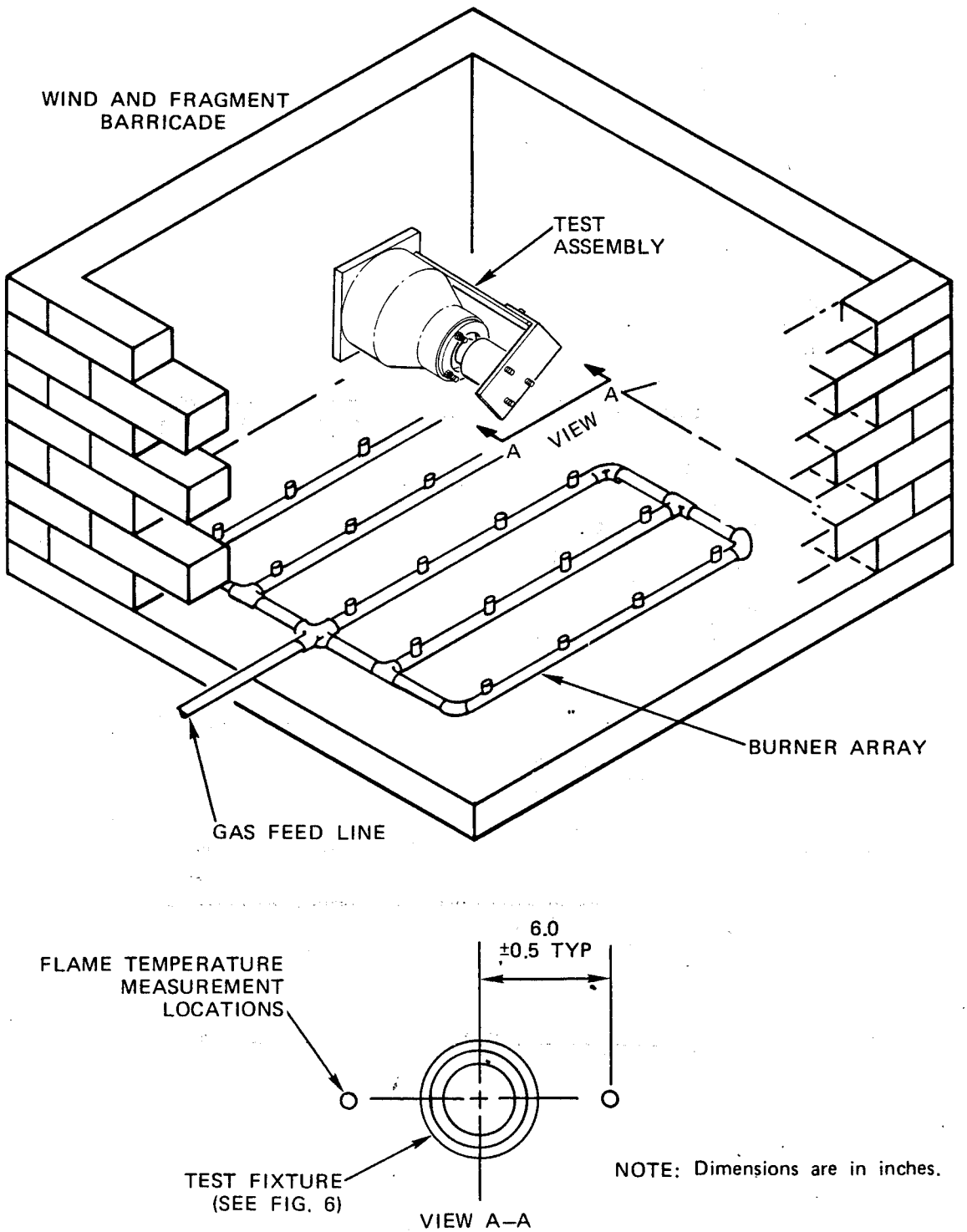


FIGURE 7. Cook-off test setup.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85599(AS)	Booster, Fuze, MARK 60 MOD 2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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